

YIXUAN LI

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Education

Southern University of Science and Technology (SUSTech)

Aug. 2022 – Jun. 2026 (Expected)

Bachelor of Science in Data Science

Shenzhen, China

- **GPA:** 3.79/4.00
- **Mathematics Courses:** Calculus, Linear Algebra, Mathematical Analysis, Discrete Mathematics and Its Applications, Probability Theory, Mathematical Statistics, Statistical Linear Models, Operational Research and Optimization
- **CS Courses:** Introduction to Computer Programming, Data Structures and Algorithm Analysis, Principles of Database Systems, Distributed Storage and Parallel Computing, Advanced Natural Language Processing, Artificial Intelligence

University of California, San Diego (UCSD)

Mar. 2025 – Jun. 2025

University and Professional Studies (UPS) Program

San Diego, CA, United States

- **GPA:** 4.00/4.00
- **Courses:** Machine Learning Algorithms, AI Search and Reasoning, Topics of Data Science: Trustworthy Machine Learning

Research Interests

- **Trustworthy Machine Learning:** Uncertainty Quantification, Hallucination Detection, Model Interpretability, LLM Alignment, Generative AI
- **AI for Science (Biomedical):** Multimodal Learning, Graph Neural Networks (GNNs), Medical Imaging

Research Experience

SUSTech Machine Learning Lab | Advisor: Prof. Hongxin Wei | SUSTech

Shenzhen, China

Project: Multimodal Learning for Spatial Transcriptomics-to-Proteomics Prediction

Main Developer

Oct. 2025 – Present

- Built an end-to-end framework for the *STP Open Challenge*, translating spatial transcriptomics into proteomic distributions on adjacent glioma tissue sections.
- Developed *UNet* and later *GAT/GCN-based graph models*, introducing a patch-based graph construction strategy to handle 160k-spot tissue sections under GPU constraints and mitigate boundary bias.
- Designed a *multimodal pipeline* integrating ResNet-derived morphological features from H&E Images with RNA embeddings, achieving **0.74 on Spearman@Top10** and currently holding **1st** position on the *public leaderboard*.
- Generated biologically coherent protein-level spatial maps that preserved tissue architecture and hotspot regions, enabling downstream analyses such as cellular microenvironment profiling.

Project: Benchmarking Uncertainty Quantification Methods in Large Language Models

Independent Researcher

Jun. 2025 – Oct. 2025

- Built a unified benchmarking pipeline for 10+ *semantic uncertainty quantification methods* (e.g., SE, DSE, SAR) on QA datasets using open-source LLMs, evaluated via AUROC, AUARC, and PRR.
- Standardized data processing and evaluation for reproducible comparison across uncertainty estimation approaches.
- Conducted an extensive literature review on *semantic uncertainty in LLMs*, connecting with hallucination detection and calibration, and summarized findings in technical reports and blog posts.

Jianqing Shi's Lab | Advisor: Prof. Jianqing Shi | SUSTech

Shenzhen, China

Project: Statistical Analysis of Mental Health Impact among Menopausal Women

Research Assistant

Sep. 2025 – Nov. 2025

- Analyzed large-scale survey data on menopausal women in Malaysia to examine the relationship between menopausal stages and mental health outcomes.
- Applied statistical methods including *Kruskal-Wallis test*, *Fisher's exact test*, *Chi-square test*, *ANCOVA*, *Cochran-Mantel-Haenszel test*, *Paired t-tests*, and *multiple imputation* for hypothesis testing and subgroup analysis.
- Identified stage-adjusted psychological risk factors and enhanced inference robustness in women's health research.
- Results reported internally due to data confidentiality.

Internship Experience

Machine Learning for Clinical Decision Support in Orthodontics

Jun. 2024 – Aug. 2024

Clinical Data Science Intern — Mentor: Dr. Ying Shan

Peking University Shenzhen Hospital

- Performed data preprocessing on clinical craniofacial imaging datasets, dental records, and treatment histories, which included cleaning, de-duplication, and standardization.
- Developed *linear and neural network models* based on craniofacial measurement variables to predict patient-specific tooth extraction plans. Achieved strong predictive performance, contributing to internal clinical adoption.

Personal Projects & Competitions

Concept Bottleneck in Generative Models: Reproduction Study | Core Member

Apr. 2025 – Jun. 2025

- Reproduced the CVPR paper on Post-hoc Concept Bottleneck frameworks: *Concept Bottleneck Autoencoder (CB-AE)*, *Concept Controller (CC)* using *StyleGAN*, replicating key quantitative and qualitative results from the original study.
- Conducted additional experiments on *Concept Extra-Interpolation*, *Entanglement*, and *Leakage* to evaluate model interpretability, robustness, and reliability under concept-level perturbations.
- Delivered a comprehensive reproduction package including *GitHub code*, *presentation slides*, and a public *technical blog*, earning **43.4/40** on the final project (**Top 1** in a UCSD class enrolling both undergraduate and graduate students).

Kaggle Competition: Fine-Tuning Gemma-2B for Chinese QA Dataset | Team Leader

Oct. 2024 – Jan. 2025

- Led a team to fine-tune *Gemma-2B* on the *GPT-4 Chinese QA dataset*. Built a complete fine-tuning pipeline covering data preprocessing, LoRA integration, and evaluation. Published notebook can be viewed *here*.
- Integrated *Retrieval-Augmented Generation (RAG)* for data augmentation and *LoRA-based PEFT* for parameter-efficient fine-tuning, resulting in improved dataset diversity, contextual grounding, and instruction-following performance.
- Evaluated model performance using *BLEU*, *ROUGE*, *METEOR*, and *BERTScore* metrics, and performed *cross-lingual robustness analysis* showing stable results across four languages.

Reinforcement Learning on FrozenLake Game Setting | Core Member

Oct. 2024 – Jan. 2025

- Implemented and evaluated the *Double DQN* and *Dueling DQN* algorithms on the FrozenLake environment, performing hyperparameter tuning and robustness analysis to achieve up to **98%** success rate in optimal path learning.
- Compiled a comprehensive report and structured a well-documented *GitHub code repository* to ensure reproducibility.

Awards

- First Class of the Merit Student Scholarship, SUSTech** (Top 5%, two-time recipient) 2023 – 2025
- Third Class of the Merit Student Scholarship, SUSTech** (Top 15%) 2022 – 2023
- Outstanding Student Award, SUSTech** (three-time recipient) 2022 – 2025
- Second Prize** China Undergraduate Mathematical Contest in Modeling (CUMCM), Guangdong Division Oct. 2024
- Honorable Mention** Mathematical Contest In Modeling (MCM) May 2024

Leadership & Activities

Community Involvement

Oct. 2023 – Present

Volunteer

SUSTech

- Taught online academic enrichment classes for students in rural areas, supporting long-term educational outreach efforts.
- Contributed over **179 hours** of volunteer service.
- Awarded **Outstanding Volunteer** honors at both the university and School of Science levels.

Shuren College Media Center

Sep. 2023 – Aug. 2024

President

SUSTech

- Led a 15-member team producing digital communications and organizing college events across official media channels.
- Received the **Excellent Student Leader Award** and the **Outstanding College Organization Award**.

Skills

- Programming:** Python, R, SQL, Java, JavaScript, HTML
- Development Tools:** PyTorch, Git, Jupyter Notebook, LaTeX, Markdown, Hadoop
- Languages:** English (TOEFL 109: R28/L27/S27/W27), Mandarin (Native), Cantonese (Fluent)